

Time dependent perturbations

Paul Haskins

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Unperturbed Hamiltonian $\hat{H}_0$ independent of t .

$$\hat{H}_0 |\psi_k^0\rangle = E_k^0 |\psi_k^0\rangle \quad (1)$$

$$(2)$$

Time dependency.

$$\hat{H}_0 |\psi(t)\rangle = i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle \quad (3)$$

Ansatz:

$$|\psi(t)\rangle = \sum_k |\psi_k^0\rangle c_k^0 e^{-i\omega_k^0 t} \quad (4)$$

with $\hbar\omega_k^0 = E_k^0$.

Time dependent perturbation: $g\hat{V}(t)$. We can have a series expansion in powers of the 'coupling constant' g .

Ansatz:

$$|\psi(t)\rangle = \sum_k |\psi_k^0\rangle (c_k^0 + g c_k^1(t) + \mathcal{O}(g^2)) e^{-i\omega_k^0 t} \quad (5)$$

Note c_k^1 is a function of t .

$$(\hat{H}_0 + g\hat{V}(t)) |\psi(t)\rangle = i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle \quad (6)$$

(not a power series)

$$(\hat{H}_0 + g\hat{V}(t)) \sum_k |\psi_k^0\rangle (c_k^0 + g c_k^1(t) + \mathcal{O}(g^2)) e^{-i\omega_k^0 t} = i\hbar \frac{\partial}{\partial t} \sum_k |\psi_k^0\rangle (c_k^0 + g c_k^1(t) + \mathcal{O}(g^2)) e^{-i\omega_k^0 t} \quad (7)$$

$$\sum_k |\psi_k^0\rangle [E_k^0 c_k^1(t) |\psi_k^0\rangle + \hat{V}(t) |\psi_k^0\rangle c_k^0] e^{-i\omega_k t} = i\hbar \sum_k |\psi_k^0\rangle \left[\frac{dc_k^1}{dt} - i\omega_k c_k^1 \right] e^{-i\omega_k t} \quad (8)$$

Where here we are only keeping first order terms.

Using our unperturbed solutions with $\langle \psi_l^0 | \star$.

$$E_l^0 c_l^1(t) + \sum_k \langle \psi_l^0 | \hat{V}(t) | \psi_k^0 \rangle c_k^0 e^{-i\omega_k t} = i\hbar \left[\frac{dc_l^1}{dt} - i\omega_l c_l^1(t) \right] e^{-i\omega_l t} \quad (9)$$

we see $E_l^0 c_l^1(t)$ cancels with $\hbar\omega_l c_l^1(t) e^{-i\omega_l t}$.

$$\frac{dc_l^1(t)}{dt} = -\frac{i}{\hbar} \sum_k \langle \psi_l^0 | \hat{V}(t) | \psi_k^0 \rangle c_k^0 e^{-i(\omega_k - \omega_l)t} \quad (10)$$

We have an example (harmonic oscillator 1D).

$$\hat{H}_0 = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{m\omega^2}{2} x^2 \quad (11)$$

If we switch on our perturbation physically we expect oscillation about the bottom of the valley. with a perturbation described by

$$g\hat{V} = g\hat{x} \sin \gamma t \tag{12}$$

and initially in the ground state.

The steps. Only non-zero value is when $k = 0$. Since starts at ground state sum disappears. Then we need to find the matrix elements. Since sum disappears the weight of the l th state decouples from other elements. All we need to know is how the potential acts on the initial state wave function. 'Infinite tower of 1st order ODEs' which are now uncoupled.